Danmarks Tekniske Universitet

02477 Bayesiansk machine learning

Aflevering 2

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May 17, 2026

1 Part 1

Task 1.1

We are given a Gaussian process, $f_i(x) \sim \mathcal{GP}(0, k_i(x, x'))$, where:

$$
\begin{aligned}
k_1(x, x') &= 2\exp\left(-\frac{(x-x')^2}{2\cdot 0.3^2}\right) \\
k_2(x, x') &= \exp\left(-\frac{(x-x')^2}{2\cdot 0.1^2}\right) \\
k_3(x, x') &= 4 + 2xx' \\
k_4(x, x') &= \exp(-2\sin(3\pi \cdot |x-x'|^2)) \\
k_5(x, x') &= \exp(-2\sin(3\pi \cdot |x-x'|^2) + 4xx' \\
k_6(x, x') &= \frac{1}{5} + \min(x, x')
\end{aligned}
$$

We want to compute the marginal prior mean and variance of the Gaussian process, i.e. $\mathbb{E}[f_i(x)]$ and $\mathbb{V}[f_i(x)]$ for each covariance function. From $f_i(x) \sim \mathcal{GP}(0, k_i(x, x'))$, we immediately see that $\mathbb{E}[f_i(x)] = m(x) = 0$ for all i. Next, we use the fact that $\mathbb{V}[f_i(x)] = \mathrm{Cov}(f_i(x), f_i(x)) = k_i(x, x)$. Therefore, we can deduce expressions for the marginal prior variance by evaluating $k_i(x, x)$:

1)

$$k_1(x, x) = 2\exp(0) = 2$$

2)

$$k_2(x, x) = \exp(0) = 1$$

3)

$$k_3(x, x) = 4 + 2x^2$$

4)

$$k_4(x, x) = \exp(-2\sin(0)^2) = \exp(0) = 1$$

5)

$$k_5(x, x) = 1 + 4x^2$$

6)

$$k_6(x, x) = \frac{1}{5} + \min(x, x) = \frac{1}{5} + x$$

Task 1.2

Which of the six covariance functions are stationary covariance functions?
Stationary covariance functions are covariance functions that depend only on the difference $h = x - x'$. From this, we can see that k_1, k_2, and k_4 are stationary. All the other covariance functions also depend on either xx' or $\min(x, x')$.

Task 1.3

We have matched the plots with the covariance functions:

- Plot A: 2
- Plot B: 6
- Plot C: 5

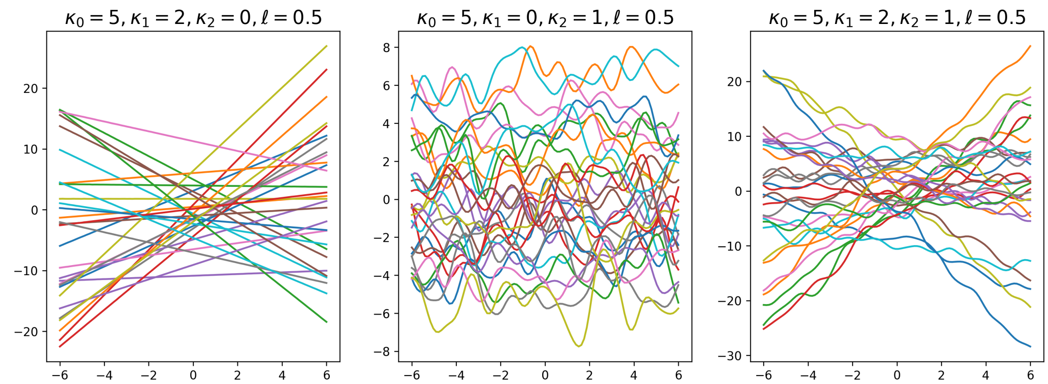

Figure 1: 30 realizations of the process $f(x) \sim \mathcal{GP}(0, k_7(x, x'))$ for three different configurations of $(\kappa_0, \kappa_1, \kappa_2, \ell)$.

- Plot D: 1
- Plot E: 4
- Plot f: 3

Task 1.4

We have sampled $S = 30$ realizations of the process $f(x) \sim \mathcal{GP}(0, k_7(x, x'))$ where:

$$k_7(x, x') = \kappa_0^2 + k_1^2 x x' + \kappa_2^2 \exp\left(-\frac{||x - x'||_2^2}{2\ell^2}\right)$$

We have repeated the experiment for three different configurations of $(\kappa_0, \kappa_1, \kappa_2, \ell)$. The results are visualized in Figure 1.
Figure 2 shows the covariance matrices as well and are made by a different student.

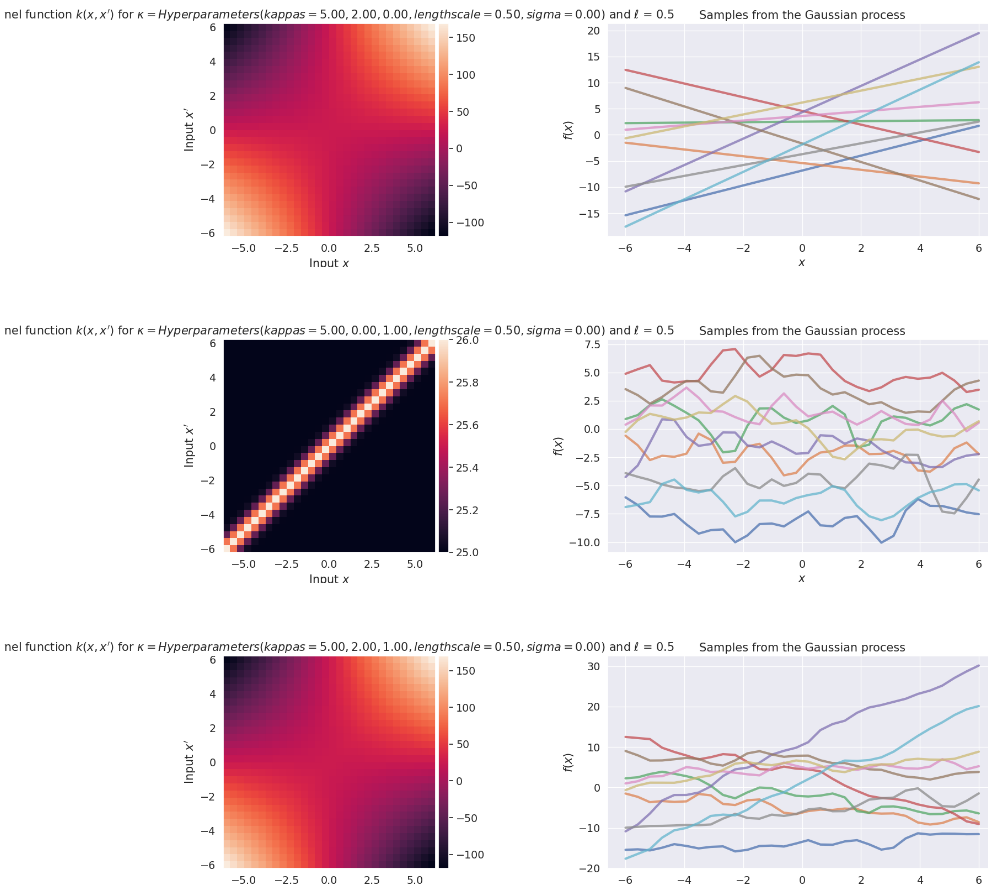

Figure 2: Visualizations of the kernel matrix and 30 realizations of the process $f(x) \sim \mathcal{GP}(0, k_7(x, x'))$ for the same three configurations of $(\kappa_0, \kappa_1, \kappa_2, \ell)$.

2 Part 2

Task 2.1

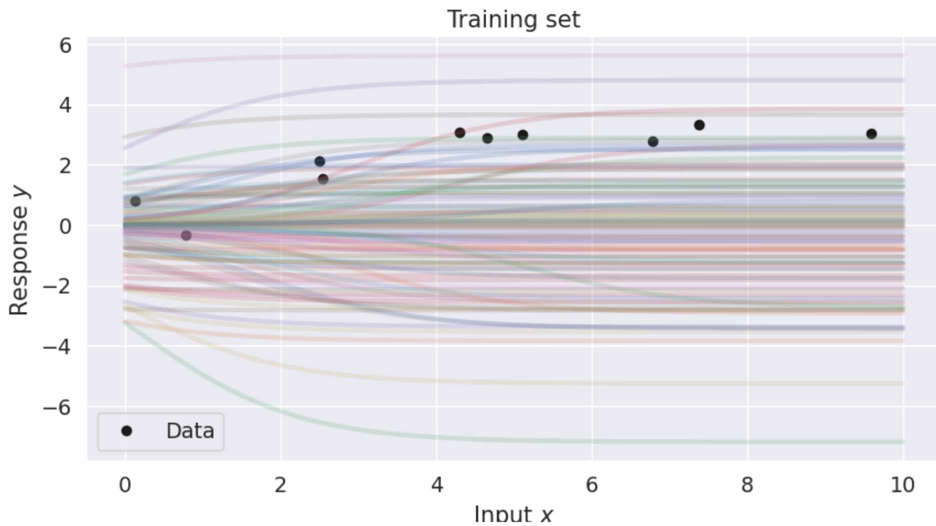

Figure 3: 100 model samples from the prior $p(w)$ plotted together with a scatter plot fo the dataset.

Task 2.2

Since the model f is non-linear due to the inclusion of the sigmoid function, the results from Murphy 3.3 on linear gaussian systems are not applicable here.

$$f(x) = w_1 \sigma(x + w_0)$$

Task 2.3

The numerical value of the log joint probability of the dataset for $\mathbf{w} = \mathbf{0}$ is -130.95. These are computed using the log normal pdf and are further simplified in this case since $f(x) = 0$ for all x when $\mathbf{w} = \mathbf{0}$.

Task 2.4

To simplify the notation and have the math derivation more closely match my code implementation, introduce the following abbreviations:

$$\begin{aligned} s_n &= \sigma(x_n + w_0) \\ s'_n &= s_n(1 - s_n) \\ s''_n &= s'_n(1 - 2s_n) \end{aligned}$$

Dropping the normalization constants, the unnormalized log joint probability is the sum of the log prior and log likelihood:

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{2\tau^2}\mathbf{w}^T\mathbf{w} - \frac{\beta}{2}\sum_{n=1}^{N}(w_1 s_n - y_n)^2$$

Taking the first partial derivatives with respect to w_0 and w_1 yields the gradient vector $\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w})$. Factoring out the term $(w_1 s_n - y_n)$, we get:

$$\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}) = -\frac{\mathbf{w}}{\tau^2} - \beta\sum_{n=1}^{N}(w_1 s_n - y_n)\begin{bmatrix} w_1 s_n' \\ s_n \end{bmatrix}$$

For the hessian, first we notice that the contribution of the prior term is simply $-\frac{1}{\tau^2}\mathbf{I}$. We calculate the contribution of the likelihood individually for each term.

First we consider the contribution of the likelihood. We start with w_0. Applying the product rule and chain rule, we get

$$\frac{\delta^2\mathcal{L}(\mathbf{w})}{\delta(w_0)^2} = \tau^{-2} - \beta\sum_{n=1}^{N}\left(w_1^2(s_n s_n'' + s_n' s_n') - w_1 y_n s_n''\right)$$

w_1 is much simpler.

$$\frac{\delta^2\mathcal{L}(\mathbf{w})}{\delta w_1 \delta w_0} = \tau^{-2} - \beta\sum_{n=1}^{N}(s_n s_n)$$

For the cross terms we get.

$$\frac{\delta^2\mathcal{L}(\mathbf{w})}{\delta(w_1)^2} = -\beta\sum_{n=1}^{N}(2w_1 s_n s_n' - y_n s_n')$$

Factoring it in like in my code implementation, we the hessian on the following form:

$$\mathbf{H} = -\frac{1}{\tau^2}\mathbf{I} - \beta\sum_{n=1}^{N}\begin{bmatrix} w_1^2(s_n')^2 + w_1(w_1 s_n - y_n)s_n'' & s_n'(2w_1 s_n - y_n) \\ s_n'(2w_1 s_n - y_n) & s_n^2 \end{bmatrix}$$

Task 2.5

We determine the Laplace approximation using the standard technique of setting $p(w|y) \approx q(w) = \mathcal{N}(w|w_{MAP}, A^{-1})$ where A is the negative Hessian found in the previous step.

Task 2.6

See figure 4.

Task 2.7

See figure 5.

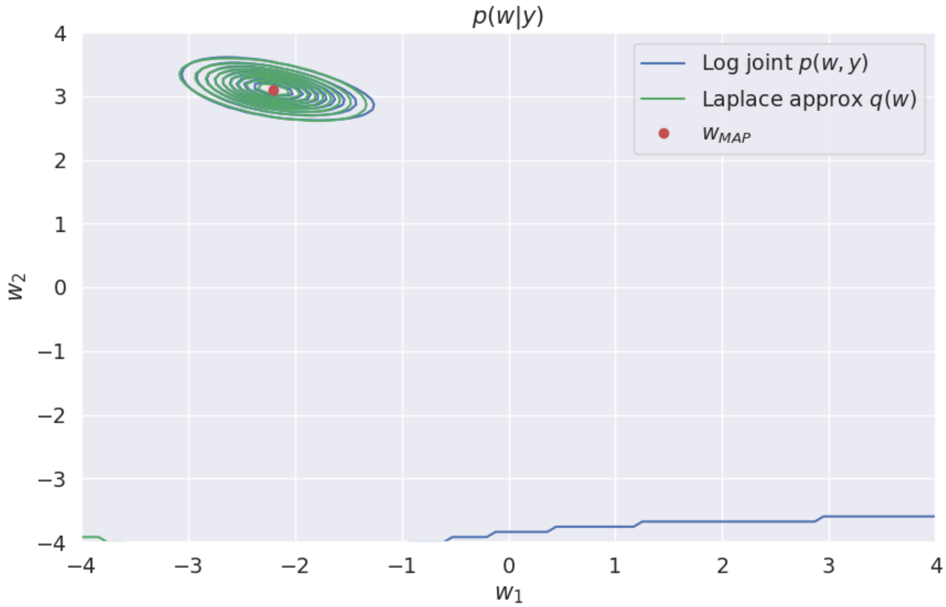

Figure 4: Contour plot of the log joint probability of the probabilistic model and the Laplace approximation along with the MAP estimate (Note that weights are 1 indexed in on the axis labels).

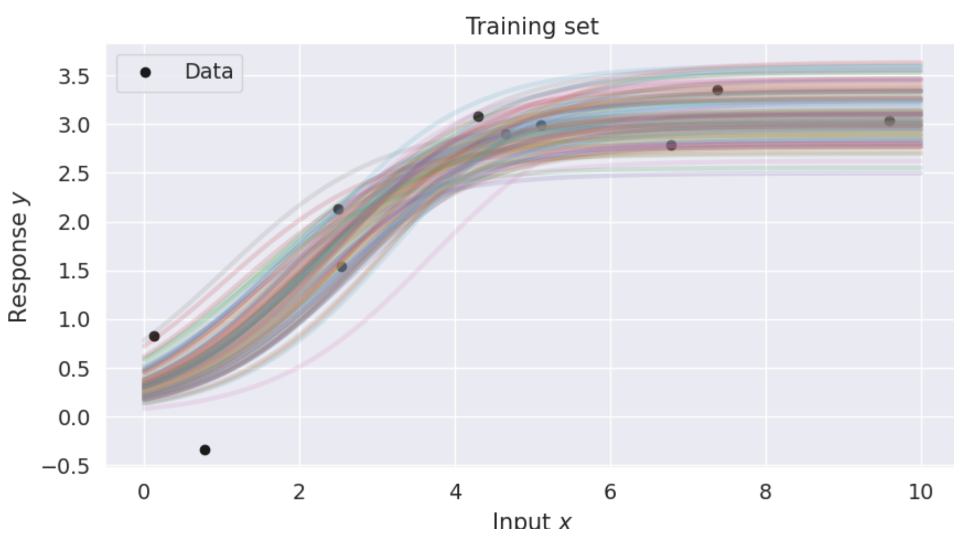

Figure 5: $S = 100$ model samples from the approximate posterior. Qualitatively, we see a much better match than with the prior.

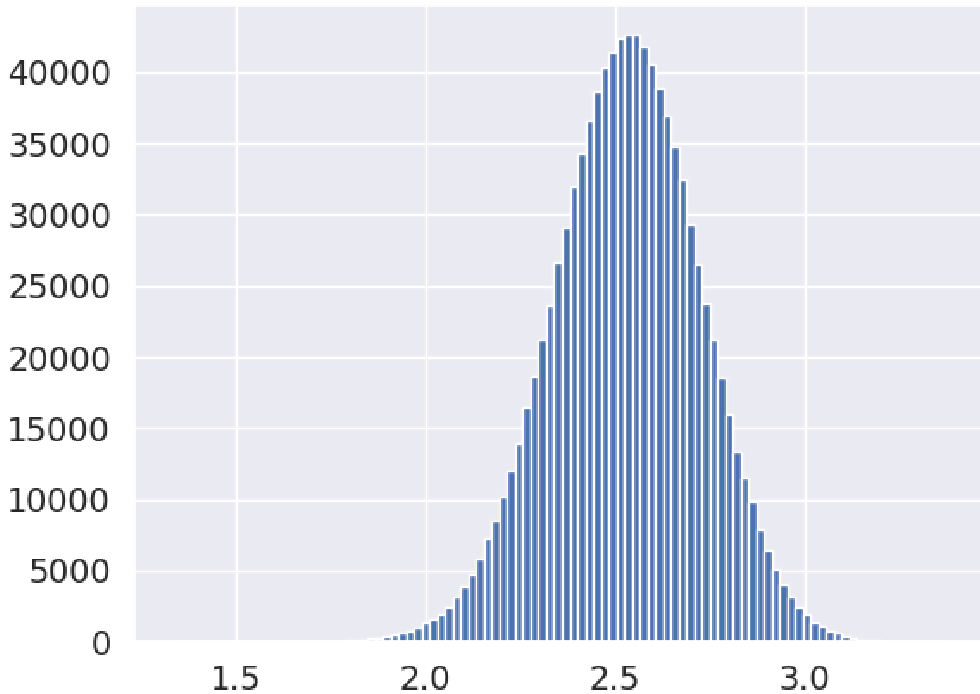

Figure 6: Histogram of empirical samples of $f(3.75)$ from the approximate posterior.

Task 2.8

We solve this problem using Monte Carlo sampling from the approximate posterior. Using 10^6 samples we find the $P(f(8) > 3) \approx 0.66$.

Task 2.9

We solve this problem in similar fashion with Monte Carlo sampling and bootstrapping. Using 10^6 samples we find $\mathbb{E}[f(3.75)] \approx 2.53$ and the 95% confidence interval to be $[2.139, 2.90]$.